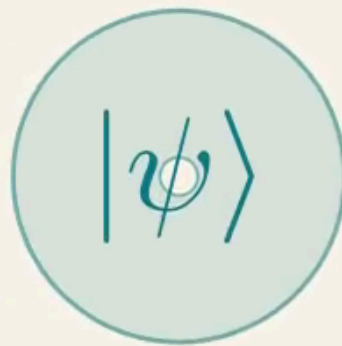




Day 4

Intro to Quantum Computing

KAUST ACADEMY + KGSP

Yesterday: three algorithms — all **fixed recipes**. Every gate predetermined.

Yesterday: three algorithms — all **fixed recipes**. Every gate predetermined.

Today: circuits with knobs.

Pop-science says

“Quantum AI will out-think ChatGPT.”

Not remotely where the field is.

Pop-science says

“Quantum AI will out-think ChatGPT.”

Not remotely where the field is.

What’s actually happening

*Small circuits with tunable knobs,
trained by an ordinary classical
optimizer.*

Promising. Unproven. Real.

Why knobs? Because of the machines we actually have.

Why knobs? Because of the machines we actually have.

Yesterday's fixed recipes assume a big, clean computer — Shor wants **millions** of near-perfect qubits.

Why knobs? Because of the machines we actually have.

Yesterday's fixed recipes assume a big, clean computer — Shor wants **millions** of near-perfect qubits.

What exists: **hundreds** of noisy ones, good for only **short** circuits.

Why knobs? Because of the machines we actually have.

Yesterday's fixed recipes assume a big, clean computer — Shor wants **millions** of near-perfect qubits.

What exists: **hundreds** of noisy ones, good for only **short** circuits.

Knobs are the adaptation: short circuits, **tuned** to squeeze the most out of imperfect hardware.

This afternoon, a quantum computer decides **where**
you sit.

This afternoon, a quantum computer decides **where**
you sit.

Input: this class's own friendship graph — who chats with whom.

Output: two project teams, printed by the last cell of today's lab.

This afternoon, a quantum computer decides **where**
you sit.

Input: this class's own friendship graph — who chats with whom.

Output: two project teams, printed by the last cell of today's lab.

Not a demo. The actual seating for the rest of the afternoon.

Today, in four moves:

A problem — split a group in two, and score the split. You'll do it by hand.

Today, in four moves:

A problem — split a group in two, and score the split. You'll do it by hand.

An engine — a circuit with knobs, and a classical coach that turns them.

Today, in four moves:

A problem — split a group in two, and score the split. You'll do it by hand.

An engine — a circuit with knobs, and a classical coach that turns them.

An algorithm — yesterday's two moves, given dials. The hero.

Today, in four moves:

A problem — split a group in two, and score the split. You'll do it by hand.

An engine — a circuit with knobs, and a classical coach that turns them.

An algorithm — yesterday's two moves, given dials. The hero.

The honest part — what this buys today, and what it doesn't (yet).

Today, in four moves:

A problem — split a group in two, and score the split. You'll do it by hand.

An engine — a circuit with knobs, and a classical coach that turns them.

An algorithm — yesterday's two moves, given dials. The hero.

The honest part — what this buys today, and what it doesn't (yet).

Problems first, names after — as always.

Splitting a group in two

One problem, many disguises.

Ten students. Some pairs **chat too much** — draw an edge between them.

Ten students. Some pairs **chat too much** — draw an edge between them.

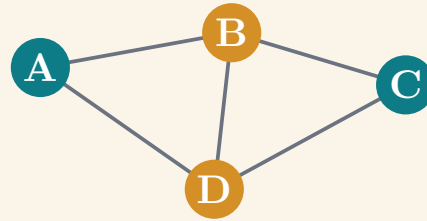
Split everyone into **two teams** so as many chatty pairs as possible end up separated.

Ten students. Some pairs **chat too much** — draw an edge between them.

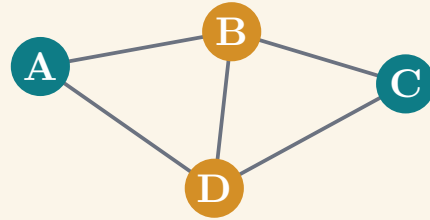
Split everyone into **two teams** so as many chatty pairs as possible end up **separated**.

Every teacher since antiquity has solved this by eye. Today: by physics.

Score a split: **count the edges whose endpoints land in different teams.**

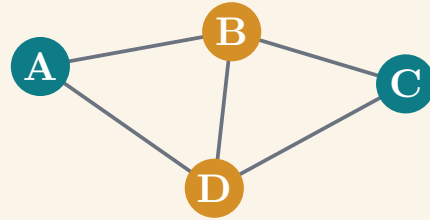


Score a split: **count the edges whose endpoints land in different teams.**



Teams $\{A, C\}$ vs $\{B, D\}$: edges A-B, B-C, A-D, D-C cross; B-D doesn't $\rightarrow$ **score 4 of 5.**

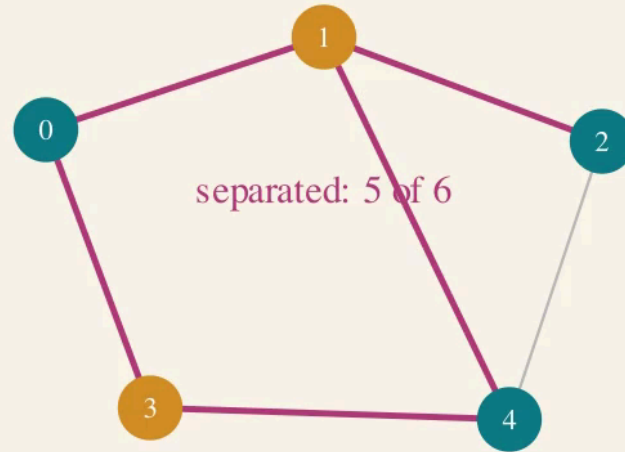
Score a split: **count the edges whose endpoints land in different teams.**



Teams $\{A, C\}$ vs $\{B, D\}$: edges A-B, B-C, A-D, D-C cross; B-D doesn't $\rightarrow$ **score 4 of 5.**

Can any split of these four do better? (Try one — this is exactly today's paper quiz.)

the goal: the split that separates the most pairs



Why not just be clever about it?

Why not just be clever about it?

By eye / greedy: move one person at a time until no move helps — gets stuck in **local traps** (a better split exists, but every **single** swap makes things worse first).

Why not just be clever about it?

By eye / greedy: move one person at a time until no move helps — gets stuck in **local traps** (a better split exists, but every **single** swap makes things worse first).

Exhaustively: guaranteed — and exponentially explosive.

Why not just be clever about it?

By eye / greedy: move one person at a time until no move helps — gets stuck in **local traps** (a better split exists, but every **single** swap makes things worse first).

Exhaustively: guaranteed — and exponentially explosive.

Hold on to “local traps” — you’ll **see** them as terrain within the hour.

Why not just try every split?

Why not just try every split?

10 people $\rightarrow 2^{10} = 1024$ splits. A laptop shrugs.

Why not just try every split?

10 people $\rightarrow 2^{10} = 1024$ splits. A laptop shrugs.

60 people $\rightarrow 2^{60} \approx 10^{18}$ splits. A data center sweats for decades.

Why not just try every split?

10 people $\rightarrow 2^{10} = 1024$ splits. A laptop shrugs.

60 people $\rightarrow 2^{60} \approx 10^{18}$ splits. A data center sweats for decades.

Any split is **easy to check**, the best one is **hard to find** — the **NP** shape from yesterday.

The puzzle you just scored has a name:

The puzzle you just scored has a name:

Max-Cut

— split the nodes to **maximize** the edges **cut**. NP-hard: no known fast exact method.

The puzzle you just scored has a name:

Max-Cut

— split the nodes to **maximize** the edges **cut**. NP-hard: no known fast exact method.

Same puzzle, other disguises: chip layout, portfolio risk clustering, community detection, timetabling.

Give every person a **qubit**. A split is then just a **bitstring** — bit = team.

Give every person a **qubit**. A split is then just a **bitstring** — bit = team.

$|0110\rangle$ = “A and D in team 0, B and C in team 1.”

Give every person a **qubit**. A split is then just a **bitstring** — bit = team.

$|0110\rangle =$ “A and D in team 0, B and C in team 1.”

So the 2^n splits **are** the 2^n basis states — and one superposition holds **all of them**.

Reading a measurement = reading a seating chart.

And we already own half the machinery:

- ✓ A superposition that holds all 2^n charts at once.

And we already own half the machinery:

- ✓ A superposition that holds all 2^n charts at once.
- ✓ Phase kickback — a gate can stamp branches without reading them.

And we already own half the machinery:

- ✓ A superposition that holds all 2^n charts at once.
- ✓ Phase kickback — a gate can stamp branches without reading them.
- ✓ Interference — mixing that turns hidden stamps into visible probability.

And we already own half the machinery:

- ✓ A superposition that holds all 2^n charts at once.
- ✓ Phase kickback — a gate can stamp branches without reading them.
- ✓ Interference — mixing that turns hidden stamps into visible probability.
- ✗ Yesterday's stamp only says **yes/no**. Splits need **how good**.

And we already own half the machinery:

- ✓ A superposition that holds all 2^n charts at once.
- ✓ Phase kickback — a gate can stamp branches without reading them.
- ✓ Interference — mixing that turns hidden stamps into visible probability.
- ✗ Yesterday's stamp only says **yes/no**. Splits need **how good**.
- ✗ And nothing so far had anything to **tune**.

Tonight's instance: **you**.

10 nodes. Edges = the friendships you report this morning (anonymously).

Tonight's instance: **you**.

10 nodes. Edges = the friendships you report this morning (anonymously).

Ten qubits, 1024 possible charts — small enough to verify classically, big enough to be yours.

A circuit with knobs

...and a classical coach that turns them.

Tuesday's lab, remember the slider? One qubit, one rotation $R_y(\theta)$, measure.

Tuesday's lab, remember the slider? One qubit, one rotation $R_y(\theta)$, measure.

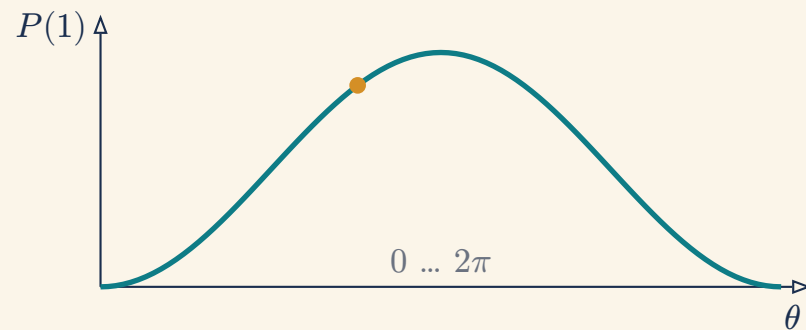
Turn θ and the outcome statistics turn **smoothly** with it: $P(1) = \sin^2(\theta/2)$.

Tuesday's lab, remember the slider? One qubit, one rotation $R_y(\theta)$, measure.

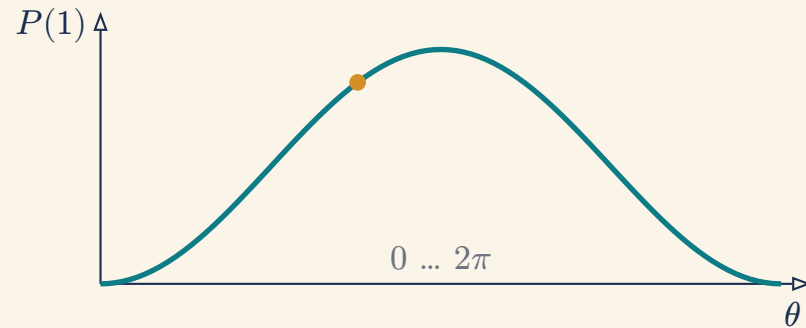
Turn θ and the outcome statistics turn **smoothly** with it: $P(1) = \sin^2(\theta/2)$.

θ is a **knob**. A gate with a dial — not a fixed move like H or X .

The knob's response, drawn:



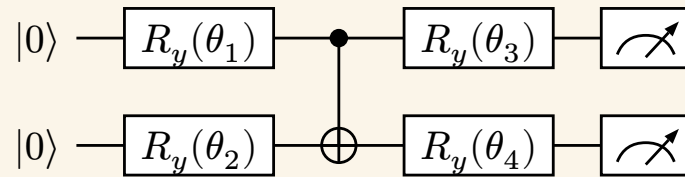
The knob's response, drawn:



No cliffs, no jumps: nudge θ and the statistics nudge back. **Smooth response is what makes tuning possible.**

Stack knobbed rotations and entanglers, and you get a circuit with a **dial panel**:

Stack knobbed rotations and entanglers, and you get a circuit with a **dial panel**:



A circuit like this is called **parameterized** — its behaviour is a function of the dial settings θ .

Run the circuit 1000 times on the Max-Cut register. Every shot returns a split — and every split has a score.

Run the circuit 1000 times on the Max-Cut register. Every shot returns a split — and every split has a score.

outcome	$ 0110\rangle$	$ 0101\rangle$	$ 0000\rangle$	$ 1111\rangle$
score	4	3	0	0
fraction	0.5	0.3	0.1	0.1

$$\text{Average score} = 0.5 \cdot 4 + 0.3 \cdot 3 + 0.2 \cdot 0 = \mathbf{2.9}$$

Run the circuit 1000 times on the Max-Cut register. Every shot returns a split — and every split has a score.

outcome	$ 0110\rangle$	$ 0101\rangle$	$ 0000\rangle$	$ 1111\rangle$
score	4	3	0	0
fraction	0.5	0.3	0.1	0.1

$$\text{Average score} = 0.5 \cdot 4 + 0.3 \cdot 3 + 0.2 \cdot 0 = \mathbf{2.9}$$

That probability-weighted average is the state's **expectation value**, written $\langle C \rangle$ — and it is the circuit's **score as a function of the knobs**.

Why score by the **average** and not just keep the best shot you saw?

Why score by the **average** and not just keep the best shot you saw?

You do keep it — the best shot is the final **answer**.

Why score by the **average** and not just keep the best shot you saw?

You do keep it — the best shot is the final **answer**.

But for **steering**, the average wins: it responds **smoothly** to the knobs, while “the best shot so far” jumps around.

Why score by the **average** and not just keep the best shot you saw?

You do keep it — the best shot is the final **answer**.

But for **steering**, the average wins: it responds **smoothly** to the knobs, while “the best shot so far” jumps around.

Coaches need smooth signals — same reason ML trains on average loss, not the single luckiest sample.

Now put a **classical optimizer** in charge of the dials:

1. Quantum: run the circuit at the current knobs, report $\langle C \rangle$.

Now put a **classical optimizer** in charge of the dials:

1. Quantum: run the circuit at the current knobs, report $\langle C \rangle$.
2. Classical: nudge the knobs to push $\langle C \rangle$ the right way.

Now put a **classical optimizer** in charge of the dials:

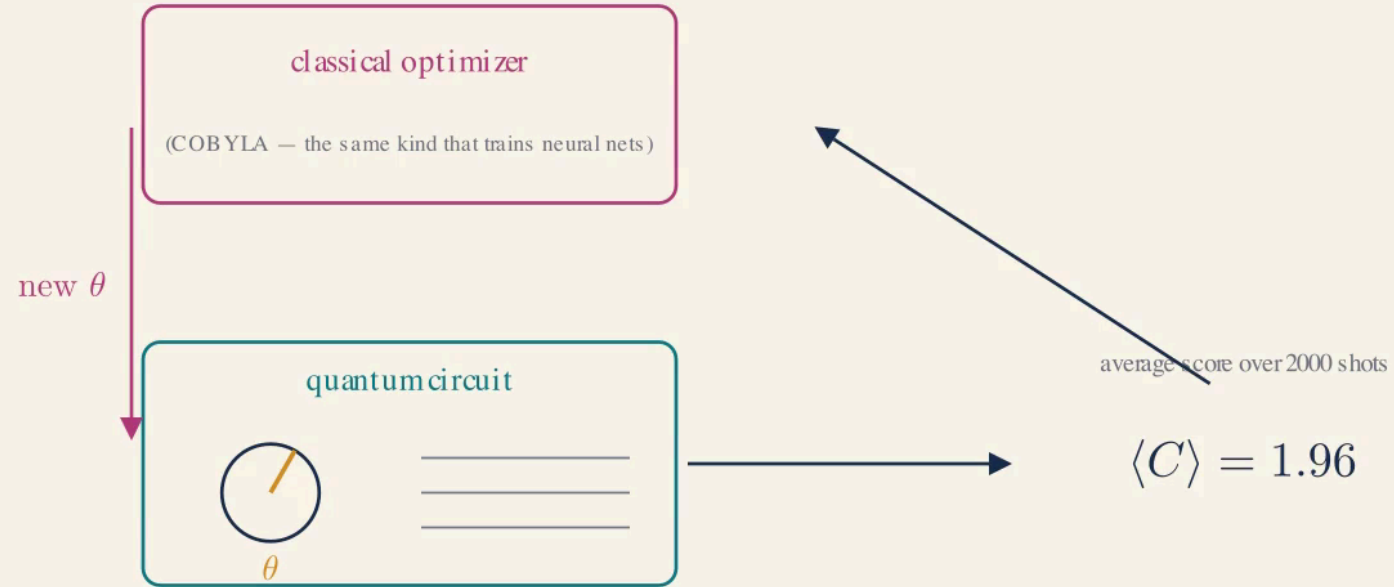
1. Quantum: run the circuit at the current knobs, report $\langle C \rangle$.
2. Classical: nudge the knobs to push $\langle C \rangle$ the right way.
3. Repeat until the score stops improving.

Now put a **classical optimizer** in charge of the dials:

1. Quantum: run the circuit at the current knobs, report $\langle C \rangle$.
2. Classical: nudge the knobs to push $\langle C \rangle$ the right way.
3. Repeat until the score stops improving.

You don't leave the quantum computer alone. You **babysit** it.

quantum evaluates — classical steers. that's the loop



Machine learning

model with weights

loss = $f(\text{weights})$

optimizer (Adam, SGD)

minibatch noise

train until loss flattens

Machine learning

model with weights
loss = $f(\text{weights})$
optimizer (Adam, SGD)
minibatch noise
train until loss flattens

This loop

circuit with knobs θ
score = $\langle C \rangle(\theta)$
optimizer (COBYLA, SPSA)
shot noise
tune until the score flattens

Where the mirror bends — three honest differences:

No backprop. You can't differentiate **through** a measurement — the coach only ever sees scores.

Where the mirror bends — three honest differences:

No backprop. You can't differentiate **through** a measurement — the coach only ever sees scores.

Every loss read costs. One glance at $\langle C \rangle$ = a thousand circuit runs.

Where the mirror bends — three honest differences:

No backprop. You can't differentiate **through** a measurement — the coach only ever sees scores.

Every loss read costs. One glance at $\langle C \rangle$ = a thousand circuit runs.

Knobs are angles. Parameters live on circles, not a line — the terrain wraps around.

“No backprop” has one beautiful loophole. Remember the knob’s response —
a **sinusoid**?

“No backprop” has one beautiful loophole. Remember the knob’s response —
a **sinusoid**?

A sinusoid’s **slope** is pinned down **exactly** by two samples a quarter-turn apart:

$$\frac{\partial \langle C \rangle}{\partial \theta} = \frac{1}{2} \left[\langle C \rangle \left(\theta + \frac{\pi}{2} \right) - \langle C \rangle \left(\theta - \frac{\pi}{2} \right) \right]$$

“No backprop” has one beautiful loophole. Remember the knob’s response —
a **sinusoid**?

A sinusoid’s **slope** is pinned down **exactly** by two samples a quarter-turn apart:

$$\frac{\partial \langle C \rangle}{\partial \theta} = \frac{1}{2} \left[\langle C \rangle \left(\theta + \frac{\pi}{2} \right) - \langle C \rangle \left(\theta - \frac{\pi}{2} \right) \right]$$

Run the **same circuit** twice at shifted knobs, subtract — an **exact gradient**, not a finite-difference guess. This move is called the **parameter-shift rule**.

“No backprop” has one beautiful loophole. Remember the knob’s response —
a **sinusoid**?

A sinusoid’s **slope** is pinned down **exactly** by two samples a quarter-turn apart:

$$\frac{\partial \langle C \rangle}{\partial \theta} = \frac{1}{2} \left[\langle C \rangle \left(\theta + \frac{\pi}{2} \right) - \langle C \rangle \left(\theta - \frac{\pi}{2} \right) \right]$$

Run the **same circuit** twice at shifted knobs, subtract — an **exact gradient**, not a finite-difference guess. This move is called the **parameter-shift rule**.

The price: one pair of runs per knob appearance, every step — gradients cost shots. (You’ll measure one in the lab’s bonus.)

Algorithms built on this loop are called

Algorithms built on this loop are called

variational quantum algorithms

- **vary** the parameters until the score stops improving. The workhorse pattern of near-term quantum computing.

Algorithms built on this loop are called

variational quantum algorithms

- **vary** the parameters until the score stops improving. The workhorse pattern of near-term quantum computing.

The quantum computer is the model. The classical computer is still the coach.

One honest wrinkle: $\langle C \rangle$ is **estimated** from finitely many shots.

One honest wrinkle: $\langle C \rangle$ is **estimated** from finitely many shots.

1000 shots $\rightarrow$ the coach reads the score through static. Same setting, slightly different number every time.

One honest wrinkle: $\langle C \rangle$ is **estimated** from finitely many shots.

1000 shots $\rightarrow$ the coach reads the score through static. Same setting, slightly different number every time.

That's why the optimizers of choice (COBYLA, SPSA) are the noise-tolerant ones — not vanilla gradient descent.

Grover's moves, given knobs

The Day-3 toolkit returns — graded.

First instinct: we own a search algorithm. Grover the 2^n charts?

First instinct: we own a search algorithm. Grover the 2^n charts?

Grover would take $\sqrt{2^n}$ steps — a square root of an exponential is **still exponential**.

First instinct: we own a search algorithm. Grover the 2^n charts?

Grover would take $\sqrt{2^n}$ steps — a square root of an exponential is **still exponential**.

And Grover's oracle only answers **is this THE one?** — nobody knows the best chart in advance to mark it.

First instinct: we own a search algorithm. Grover the 2^n charts?

Grover would take $\sqrt{2^n}$ steps — a square root of an exponential is **still exponential**.

And Grover's oracle only answers **is this THE one?** — nobody knows the best chart in advance to mark it.

Today's bet is different: **exploit the scores**. Grade, don't gatekeep.

Yesterday's engine: the **oracle** stamped -1 on **the** answer; **diffusion** mixed the branches until it dominated.

Yesterday's engine: the **oracle** stamped -1 on **the** answer; **diffusion** mixed the branches until it dominated.

But that stamp is **all-or-nothing** — right answer or wrong answer.

Yesterday's engine: the **oracle** stamped -1 on **the** answer; **diffusion** mixed the branches until it dominated.

But that stamp is **all-or-nothing** — right answer or wrong answer.

Splits aren't right or wrong. They have **scores**. We need a stamp with a **dial**.

Move 1 — the graded stamp. Turn each split's phase in proportion to its score:

$$|\text{split}\rangle \rightarrow e^{-i \gamma \cdot \text{score}(\text{split})} |\text{split}\rangle$$

Move 1 — the graded stamp. Turn each split's phase in proportion to its score:

$$|\text{split}\rangle \rightarrow e^{-i \gamma \cdot \text{score}(\text{split})} |\text{split}\rangle$$

Good splits get stamped harder. The knob γ sets **how hard the stamp presses**.

Move 1 — the graded stamp. Turn each split's phase in proportion to its score:

$$|\text{split}\rangle \rightarrow e^{-i \gamma \cdot \text{score}(\text{split})} |\text{split}\rangle$$

Good splits get stamped harder. The knob γ sets **how hard the stamp presses**.

Phase kickback builds it, exactly like yesterday's oracle — one small gate per edge of the graph.

Move 2 — the gentle mixer. Rotate every qubit a little (R_x by the knob β):

Move 2 — the gentle mixer. Rotate every qubit a little (R_x by the knob β):

The rotation makes neighbouring splits' branches **interfere** — the stamped phases turn into probability shifts.

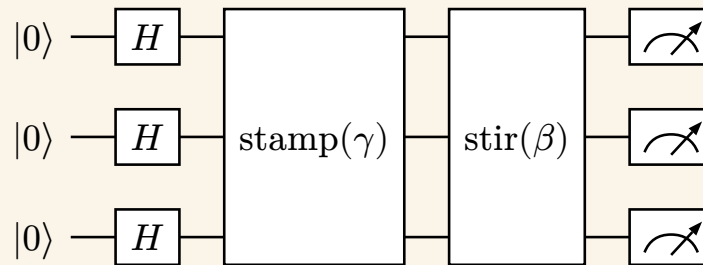
Move 2 — the gentle mixer. Rotate every qubit a little (R_x by the knob β):

The rotation makes neighbouring splits' branches **interfere** — the stamped phases turn into probability shifts.

It's the diffusion idea with a **volume knob**: β sets how much to stir.

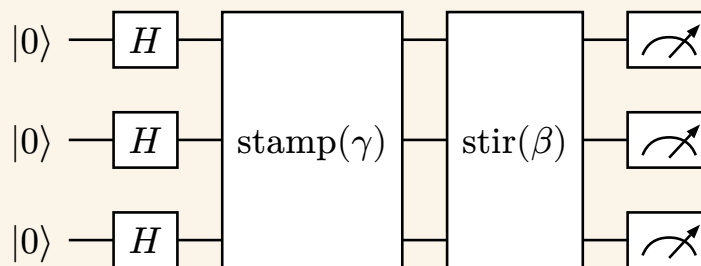
The whole recipe:

The whole recipe:



H 's put every split in superposition; then the (stamp, stir) block — possibly repeated p times — then measure.

The whole recipe:



H 's put every split in superposition; then the (stamp, stir) block — possibly repeated p times — then measure.

At depth p : the knob panel is $(\gamma_1, \beta_1, \dots, \gamma_p, \beta_p)$ — **2p dials** for the coach to tune.

What repeating the block actually does:

What repeating the block actually does:

One (stamp, stir) round = one **gentle nudge** of the whole superposition toward better charts.

What repeating the block actually does:

One (stamp, stir) round = one **gentle nudge** of the whole superposition toward better charts.

p rounds = a gradual descent toward good answers, **chopped into steps you can tune.**

What repeating the block actually does:

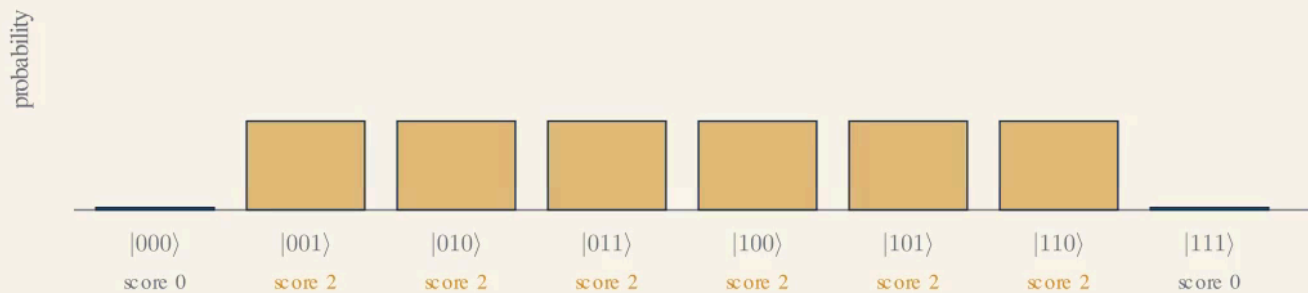
One (stamp, stir) round = one **gentle nudge** of the whole superposition toward better charts.

p rounds = a gradual descent toward good answers, **chopped into steps you can tune**.

Theory result worth knowing: infinitely many, infinitely gentle steps ($p \rightarrow \infty$) provably reach the exact optimum. Real machines can't afford that — hence the knobs.

one more (stamp, mix) layer — $p = 2$ sharpens it further

average score 2.00 of 2 — measuring now returns a best cut essentially every time



Each layer has two knobs:

Each layer has two knobs:

γ — **push toward good answers.** (Stamp too hard and the phases wrap past useful.)

Each layer has two knobs:

γ — **push toward good answers.** (Stamp too hard and the phases wrap past useful.)

β — **keep exploring.** (Stir too much and you scramble what the stamp built.)

Each layer has two knobs:

γ — **push toward good answers.** (Stamp too hard and the phases wrap past useful.)

β — **keep exploring.** (Stir too much and you scramble what the stamp built.)

Neither knob has a formula. **Finding the right settings is the whole game** — that's what the coach is for.

Fix $p = 1$. Then the score $\langle C \rangle(\gamma, \beta)$ is a **terrain** over two knobs.

Fix $p = 1$. Then the score $\langle C \rangle(\gamma, \beta)$ is a **terrain** over two knobs.

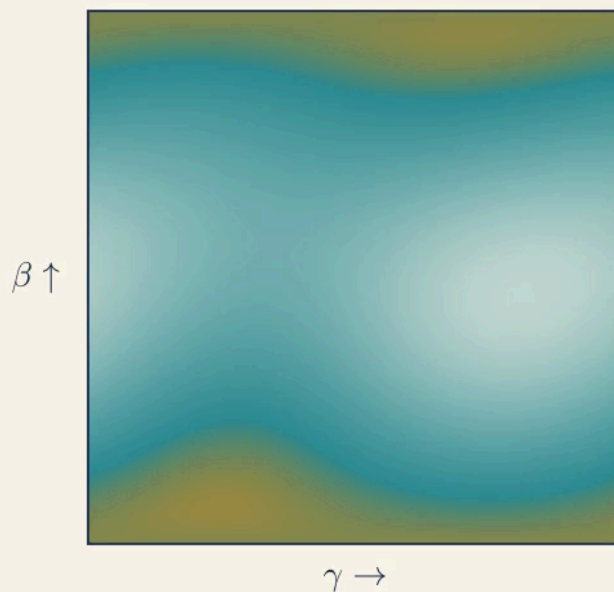
Every point of the map = one dial setting. Bright = high average score.

Fix $p = 1$. Then the score $\langle C \rangle(\gamma, \beta)$ is a **terrain** over two knobs.

Every point of the map = one dial setting. Bright = high average score.

Optimizing = **hiking this terrain** — and the coach can only feel the ground under its feet.

add a second layer ($p = 2$): higher peaks — rockier terrain



bright = high average score

amber = the peak

best now 3.82 of 4 — rockier, too

One honest read of that map: it has **false peaks**.

One honest read of that map: it has **false peaks**.

A blind hiker who only ever climbs can top out on a ridge that isn't the summit — the **local traps** from the by-eye method, now visible as terrain.

One honest read of that map: it has **false peaks**.

A blind hiker who only ever climbs can top out on a ridge that isn't the summit — the **local traps** from the by-eye method, now visible as terrain.

Working defenses: **restarts** from several places, and **warm starts** from knob settings that worked on similar graphs.

One honest read of that map: it has **false peaks**.

A blind hiker who only ever climbs can top out on a ridge that isn't the summit — the **local traps** from the by-eye method, now visible as terrain.

Working defenses: **restarts** from several places, and **warm starts** from knob settings that worked on similar graphs.

The lab hands you both tricks — watch COBYLA land somewhere mediocre and rescue it.

The recipe you've been watching has a name:

The recipe you've been watching has a name:

QAOA — Quantum Approximate Optimization Algorithm

— Farhi, Goldstone & Gutmann, 2014.

The recipe you've been watching has a name:

QAOA — Quantum Approximate Optimization Algorithm

— Farhi, Goldstone & Gutmann, 2014.

Read it honestly: **Approximate** is in the name. It aims for **good** splits, tuned —
not guaranteed ones.

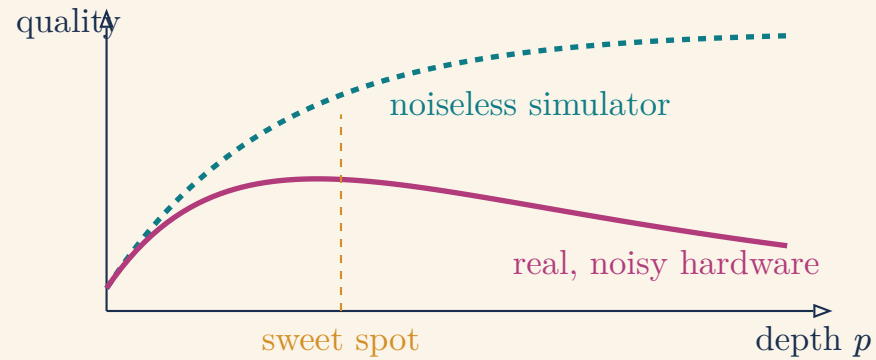
At $p = 2$ the peak was higher than at $p = 1$.

At $p = 2$ the peak was higher than at $p = 1$.

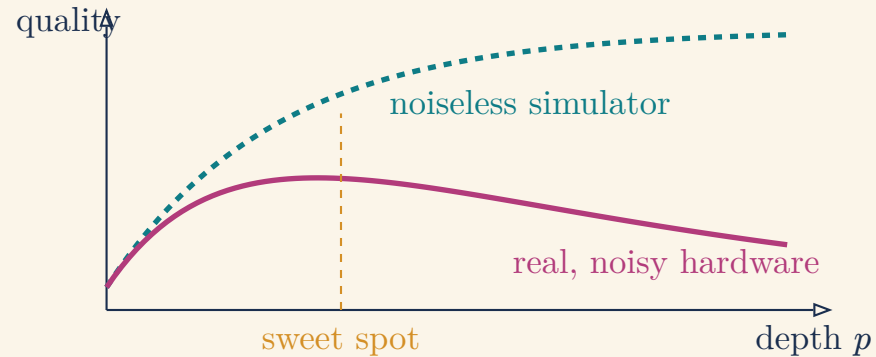
Predict: crank p to 100.

Much better, a little better — or worse? (show of hands)

In **theory**, deeper never hurts ($p \rightarrow \infty$ becomes exact). In **practice**:



In **theory**, deeper never hurts ($p \rightarrow \infty$ becomes exact). In **practice**:



Every layer adds gates $\rightarrow$ noise compounds; and $2p$ knobs $\rightarrow$ the coach gets lost. **More is not automatically better — again.**

Notice the pattern — **third time this week:**

Grover iterations — rotate past $|s\rangle$ and the odds **fall**.

Notice the pattern — **third time this week:**

Grover iterations — rotate past $|s\rangle$ and the odds **fall**.

The stamp and the stir — press or mix too hard and you scramble the gains.

Notice the pattern — **third time this week:**

Grover iterations — rotate past $|s\rangle$ and the odds **fall**.

The stamp and the stir — press or mix too hard and you scramble the gains.

QAOA depth — deeper drowns in noise and lost coaches.

Notice the pattern — **third time this week:**

Grover iterations — rotate past $|s\rangle$ and the odds **fall**.

The stamp and the stir — press or mix too hard and you scramble the gains.

QAOA depth — deeper drowns in noise and lost coaches.

Quantum resources have sweet spots, not “more is better” dials.

The honest scorecard:

No proven speedup. Unlike Grover's guaranteed $\sqrt{N}$, QAOA's advantage is an open question.

The honest scorecard:

No proven speedup. Unlike Grover's guaranteed $\sqrt{N}$, QAOA's advantage is an open question.

But it fits the machines we have. Short, shallow, noise-tolerant-ish circuits — built for today's small, noisy devices (the era the field calls NISQ).

The honest scorecard:

No proven speedup. Unlike Grover's guaranteed $\sqrt{N}$, QAOA's advantage is an open question.

But it fits the machines we have. Short, shallow, noise-tolerant-ish circuits — built for today's small, noisy devices (the era the field calls NISQ).

That's the bet: usable answers now, on imperfect hardware — while the fault-tolerant machines Shor needs are still being built.

Same loop, other costumes

Swap the cost: instead of a cut score, measure a **molecule's energy**.

Swap the cost: instead of a cut score, measure a **molecule's energy**.

Electrons settle into their lowest-energy arrangement — and that arrangement **is** the molecule's chemistry: how it binds, reacts, catalyzes.

Swap the cost: instead of a cut score, measure a **molecule's energy**.

Electrons settle into their lowest-energy arrangement — and that arrangement **is** the molecule's chemistry: how it binds, reacts, catalyzes.

Predict ground-state energies and you can hunt **drugs, catalysts, battery materials** — before touching a beaker.

Swap the cost: instead of a cut score, measure a **molecule's energy**.

Electrons settle into their lowest-energy arrangement — and that arrangement **is** the molecule's chemistry: how it binds, reacts, catalyzes.

Predict ground-state energies and you can hunt **drugs, catalysts, battery materials** — before touching a beaker.

Simulating electrons classically explodes with molecule size — this is quantum computing's original home turf (Feynman's pitch, 1981).

The same loop, in a lab coat:

The same loop, in a lab coat:

The circuit's knobs shape a guess at the electron arrangement; the measured cost is its **energy** the coach turns knobs to **lower** it.

The same loop, in a lab coat:

The circuit's knobs shape a guess at the electron arrangement; the measured cost is its **energy** the coach turns knobs to **lower** it.

This costume is called **VQE** — the **variational quantum eigensolver**. The industry workhorse beside QAOA.

The same loop, in a lab coat:

The circuit's knobs shape a guess at the electron arrangement; the measured cost is its **energy** the coach turns knobs to **lower** it.

This costume is called **VQE** — the **variational quantum eigensolver**. The industry workhorse beside QAOA.

Tomorrow's lab has a VQE track: the hydrogen molecule, for real.

Costume two: drop the circuit **inside** an ordinary neural network — as **one layer**.

Costume two: drop the circuit **inside** an ordinary neural network — as **one layer**.

classical layers → **quantum layer** → classical layers → prediction

(the incoming numbers set the circuit's angles; its measured statistics feed the next layer)

Costume two: drop the circuit **inside** an ordinary neural network — as **one layer**.

classical layers → **quantum layer** → classical layers → prediction

(the incoming numbers set the circuit's angles; its measured statistics feed the next layer)

Trained **end-to-end**: ordinary backprop outside — the **parameter-shift rule** carries the gradient **through** the quantum middle.

Costume two: drop the circuit **inside** an ordinary neural network — as **one layer**.

classical layers → **quantum layer** → classical layers → prediction

(the incoming numbers set the circuit's angles; its measured statistics feed the next layer)

Trained **end-to-end**: ordinary backprop outside — the **parameter-shift rule** carries the gradient **through** the quantum middle.

Networks built this way are called **hybrid quantum-classical nets** — the circuit as one exotic, hard-to-imitate layer.

Costume three: don't model at all — use the circuit only to judge **how similar two data points are**.

Costume three: don't model at all — use the circuit only to judge **how similar two data points are**.

Encode each data point as a quantum state (angles again). The **overlap** of two states — measurable — is their similarity score.

Costume three: don't model at all — use the circuit only to judge **how similar two data points are**.

Encode each data point as a quantum state (angles again). The **overlap** of two states — measurable — is their similarity score.

Hand the table of similarities to a plain classical SVM. The quantum computer never classifies — it only **compares**.

Costume three: don't model at all — use the circuit only to judge **how similar two data points are**.

Encode each data point as a quantum state (angles again). The **overlap** of two states — measurable — is their similarity score.

Hand the table of similarities to a plain classical SVM. The quantum computer never classifies — it only **compares**.

This pattern is a **quantum kernel method**. The bet: quantum similarity landscapes that classical features can't fake. Jury's out — but it's the cleanest-posed question in QML.

So when a headline says “*quantum machine learning*”...

So when a headline says “*quantum machine learning*”...

...it almost always means one of the three costumes you now own: the **variational loop**, a **quantum layer** in a classical net, or a **quantum kernel**.

So when a headline says “*quantum machine learning*”...

...it almost always means one of the three costumes you now own: the **variational loop**, a **quantum layer** in a classical net, or a **quantum kernel**.

Whether any of them beats a classical net at scale is genuinely unknown. Now you can read those headlines with the right filter.

So: which of this week's ideas survive contact with reality?

So: which of this week's ideas survive contact with reality?

Tomorrow: the honest tour — chemistry, finance, logistics — what's real, what's hype, and how to keep building after this week (the quantum cloud).

Where we are in the week:

Day 1 — the rules of the game (superposition, measurement, entanglement).

Where we are in the week:

Day 1 — the rules of the game (superposition, measurement, entanglement).

Day 2 — the math that runs it (states, gates, the Born rule).

Where we are in the week:

Day 1 — the rules of the game (superposition, measurement, entanglement).

Day 2 — the math that runs it (states, gates, the Born rule).

Day 3 — why it can be fast (interference, Grover, Shor).

Where we are in the week:

Day 1 — the rules of the game (superposition, measurement, entanglement).

Day 2 — the math that runs it (states, gates, the Born rule).

Day 3 — why it can be fast (interference, Grover, Shor).

Day 4 — how it's tuned when hardware is small (the variational loop, QAOA).

Where we are in the week:

Day 1 — the rules of the game (superposition, measurement, entanglement).

Day 2 — the math that runs it (states, gates, the Born rule).

Day 3 — why it can be fast (interference, Grover, Shor).

Day 4 — how it's tuned when hardware is small (the variational loop, QAOA).

Day 5 — what's real, what's hype, and where you go next.

Today, in four lines:

Max-Cut — split a group, score = separated pairs; easy to check, NP-hard to solve.

Today, in four lines:

Max-Cut — split a group, score = separated pairs; easy to check, NP-hard to solve.

Expectation value — probability-weighted average of the measured scores; the circuit's loss.

Today, in four lines:

Max-Cut — split a group, score = separated pairs; easy to check, NP-hard to solve.

Expectation value — probability-weighted average of the measured scores; the circuit's loss.

Variational — quantum evaluates, a classical coach turns the knobs.

Today, in four lines:

Max-Cut — split a group, score = separated pairs; easy to check, NP-hard to solve.

Expectation value — probability-weighted average of the measured scores; the circuit's loss.

Variational — quantum evaluates, a classical coach turns the knobs.

QAOA — the graded stamp + the gentle mixer, $\times p$, tuned.

Approximate by design.

After lunch you'll tune QAOA — and it will seat you.

Baseline: score cuts by hand and by code on a 4-node graph.

Stretch: build the mixer, assemble QAOA, find the depth sweet spot.

Aspirational: the class friendship graph — the last cell prints
the seating chart you'll actually sit in.

After lunch

- ✓ Open today's lab on MOLAB (link posted)
- ✓ Give it a moment — it auto-installs Qiskit + SciPy
- ✓ Friendship pairs: submit the form before you leave

Snagged? grab a TA.
Setup stays mechanical.